FAST- National University of Computer & Emerging Sciences, Karachi.
School of Computing,
Final Examination, Spring 2023
22nd May 2023, 12:00 pm – 03:00 pm

Course Code: CS-2001	**Course Name:** Data Structures
Instructors: Anam Qureshi, Safia Baloch, Nida Munawwar, and Mafaza Mohi	
Student Roll No:	**Section:**

Instructions:

- Except your Roll No and Section, DO NOT WRITE anything on this paper.
- Return the question paper with your answer sheet.
- Read each question completely before answering it. There are 7 **questions on 3 pages.**
- In case of any ambiguity, you may make assumptions but your assumption must not contradict any statement in the question paper.
- All the answers must be solved according to the SEQUENCE given in the question paper, otherwise points will be deducted.

Time Allowed: 180 minutes **Maximum Points**: 50

Graph Data Structure

Question No. 1 [CLO 4] **[Points: 8]**

Suppose you are working for a construction company that has been tasked with building a new road network in a remote area. The company wants to build the road network to minimize the total cost while ensuring that all areas are connected to the network. To accomplish this, they want you to design a solution that will provide them with minimum cost. Apply the proposed solution to the given adjacency matrix.

	A	B	C	D	E
A	0	3	1	∞	∞
B	3	0	5	6	7
C	1	5	0	2	4
D	∞	6	2	0	8
E	∞	7	4	8	0

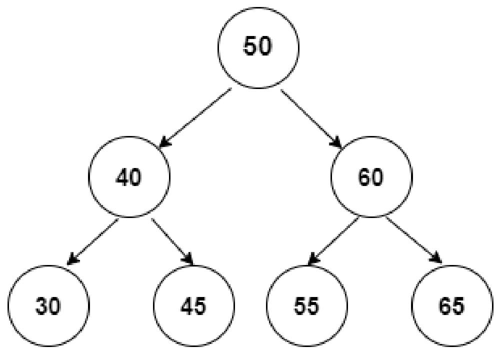
Hashing and Balanced Binary Search Tree

Question No. 2 [CLO 3] **[Points: 7]**

Grace is implementing a program for the popular word game Scrabble. She is considering the use of a hashtable (using separate chaining) or a balanced binary search tree to store the dictionary of legal words with their definitions. For the hash table, she would use a hash function that adds up the ASCII values of all of the letters in the word. Whereas, the balanced binary search tree is ordered by alphabetical ordering. Which data structure, hash table or balanced binary search tree, would allow her to find all words that contain AERST once more easily? Explain your choice. Also, write an algorithm for the justification of your choice.

Binary Search Tree	
Question No. 3 [CLO 2]	**[Points:7]**

Ahmed and Ali are playing a unique game, there are seven balls on the snooker table, and each opponent has consecutive three turns to hit the balls forming the largest trio. It's Ali's turn to have three consecutive turns. Write a program that will help him to find the largest trio of balls. The balls are arranged in the following manner.



Binary Heap/ Priority Queue	
Question No. 4 [CLO 1]	**[Points:7]**

You are building a call center application that handles customer calls. Each call has a priority assigned to it based on its importance. Design a priority queue to efficiently manage incoming calls and handle them based on priority. Implement the following operations:
Enqueue_call (call, priority): Add a call with the given priority to the queue
Dequeue_call (): Retrieve and remove the highest priority call from the queue

Stack and Queues	
Question No. 5 [CLO 1]	**[Points:7]**

Design a program that collects daily price quotes for some stock and returns the span of the stock's price for the current day. The span of the stock's price in one day is the maximum number of consecutive days (starting from that day and going backward) for which the stock price was less than or equal to the price of that day.
For example, if the prices of the stock in the last four days is [7,2,1,2] and the price of the stock today is 2, then the span of today is 4 because starting from today, the price of the stock was less than or equal to 2 for 4 consecutive days.
Also, if the prices of the stock in the last four days is [7,34,1,2] and the price of the stock today is, then the span of today is 3 because starting from today, the price of the stock was less than or equal to 8 for 3 consecutive days.

Advanced Sorting Techniques and Linked List	
Question No. 6 [CLO 3]	**[Points:7]**

You work for a social media company that allows users to create and share playlists of songs. Each playlist is represented as a linked list of songs, where each song is represented by a node in the linked list. Users can follow other users and view their playlists. To improve the user experience, the company wants to implement a feature that merges two or more playlists of a user and his followed users into one playlist that is sorted by the song title.

Write a program that takes an array of linked lists representing the playlists to merge and return a sorted linked list containing all the songs from the original playlists. Each song is represented by a node in the linked list with the following properties: song title, artist name, and duration in seconds. The linked list should be sorted by song title in ascending order.

Recursion and Backtracking	
Question No. 7 [CLO 2]	**[Points:7]**

You are given an integer array of cookies, where cookies[i] denotes the number of cookies in the i^{th} bag. You are also given an integer k that denotes the number of children to distribute all the bags of cookies to. All the cookies in the same bag must go to the same child and cannot be split up.

The unfairness of a distribution is defined as the maximum number of total cookies obtained by a single child in the distribution. Write a recursive program that will return the minimum unfairness of all distributions.

For Example:

Input: cookies = [8,15,10,20,8], k=2

Output: 31

Explanation: One optimal distribution is [8,15,8] and [10,20]

- The first child receives [8,15,8] which has a total of 8+15+8=31 cookies
- The second child receives [10,20] which has a total of 10+20=30 cookies

The unfairness of the distribution is max (31,30) =31.

******************** Good Luck ********************